

Omnivise iSER® Actuator

Intelligent Self-Contained Electrohydraulic Redundant Actuator

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Overview

These patented actuators were developed to precisely actuate steam admission valves but can be applied to any load moving, positioning, and holding application. Because they were developed for the power generation and oil and gas industries, they meet the most stringent requirements for safety, reliability, and ruggedness.

Operation

The modulating actuator precisely controls the position of steam valves, typically located near or on the steam chest assembly.

These actuator assemblies are double-acting. Motors and pumps provide hydraulic pressure to either side of the piston and check valves maintain the hydraulic pressure required to hold the piston in place when a desired position has been reached.

LVDTs provide position feedback for extremely accurate closed-loop control and near infinite positioning.

The actuator also relies on the return spring as a means for fail-safe protection. For turbine protection, the actuator is designed with a trip function to rapidly close the valve. The trip function is accomplished through the porting of high-pressure oil from the actuator, thus allowing spring force to rapidly close the valve.

Component parts

The modulating actuator assembly is comprised of the following components:

- High-pressure actuator assembly
- Redundant motor/pump assemblies (may be ordered with simplex or triplex motor/pumps)
- Control and trip manifolds with integral components (different options available)
- Remote control panel with emergency stop button and HMI display
- Duplex LVDTs for feedback (may be ordered with simplex or triplex)
- Return spring for fail-safe shut down
- Mounting hardware and connection shaft

Advantages at-a-glance

- Extremely accurate positioning
- Avoids major oil leaks
- No expensive hydraulic power unit needed
- Reduces energy consumption
- Actuator still operational if one pump, motor, LVDT or trip solenoid fails
- Online diagnostics
- Early warning of wear and leakage
- Industry 4.0 ready
- Easy connectivity to DCS or TCS
- Fail-safe operation
- Fully assembled and tested in our factory
- Standard components, lowering costs
- Eliminates piping installation
- Eliminates servo valves
- Actuator oil volume is less
- Eliminates use of fire-resistant fluids
- Large real estate savings
- Quick installation

Actuator Hardware

Hydraulic Cylinder

The hydraulic cylinder is specially designed for high pressures and special seals for preventing notable leakage.

Control Manifold

Houses all of the pumps, valves, and sensors for operation. The only external fitting is for the makeup reservoir and filters, if used.

Trip Manifold

Houses all solenoid valves and items needed for online valve stroke testing. Trip manifold can be configured to suit customers reliability and redundancy needs.

LVDT Feedback

Equipped with rugged LVDTs designed to meet the demanding applications of the power generation industry. Simplex, duplex, or triplex configurations are available as well as integral LVDTs which mount inside the hydraulic cylinder.

Spring Return

Spring is installed to operate as a fail-safe mechanism to close the steam valve in the event of an emergency. Custom configurations for fail-safe operation available.

Actuator Specifications

Temperature Range:
-4 °F to 180°F (-20°C to 82°C)
Fluid Cleanliness ISO Code: 18-16-13
Trip Time: 200 ms (configuration dependent)
Standard seals are FKM/Viton

Hydraulic Fluids:

Must be non water-based
Must be non abrasive
Must be clean prior to filling
Must match desired operating temperature range

Remote Control Panel Hardware

Power Supply

Integral 24VDC power supply for PLC power and control functions

PLC

The PLC is the newest technology of Siemens PLC incorporating integrated I/O, compact design, and function block based programming.

HMI

Rugged outdoor HMI to display testing routines, configuration settings, diagnostic information, calibration mode and manual mode.

Emergency Trip Pushbutton

Local pushbutton trips actuator in event of an emergency.

Integral Components

Circuit breakers and fuses for isolation and safety.

Control Panel Specifications

NEMA 4 (Other ratings available)
Temperature Range:
32-122°F (0-50°C)

Inputs

Power: 125 VDC 12A (AC versions available)
Trip Input(s): 24 VDC, 120 VAC, or 125 VDC
Position Demand Signal: 4-20 mA
Profinet or ModbusTCP Communications

Outputs

Actuator Position: 4-20 mA
Actuator or Motor Temperature: 4-20 mA
Actuator Open: Digital contact
Actuator Closed: Digital contact
Actuator Tripped: Digital contact
Alarm Output: Digital contact
Warning Output: Digital contact
Power Supply Fault: Digital Contact
Profinet or ModbusTCP Communications

Options

Seal Options

- Nitrile/Buna-N
- Fluorocarbon/Viton

Rating Options

- Various NEMA ratings available
- Class I Division 2 Groups (pending)
- SIL 3 (pending)
- API 612 trip manifold

Actuator Temperature Options

- Lower -30°F (-34°C)
- Higher 212°F (100°C)
- Special applications available upon request

Pushbutton Options

Local push buttons can be customized to customer needs.

Control Panel Options

Addition DI/DO/AO/AI available
Redundant power supplies
Redundant PLCs

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